IMERYS
Metalcasting Solutions

Greg

SAFETY DATA SHEET

Product Trade Name: **PREMIX™**

Revision Date: 29-MAY-2015

1. CHEMICAL PRODUCT AND COMPANY IDENTIFICATION

Product Trade Name:	Premix™
Synonyms:	None
Chemical Family:	Mineral
Application:	Additive
Manufacturer/Supplier	S&B Industrial Minerals N.A., Inc. 920 Cassatt Rd. Suite 205 Berwyn, PA. 19312 Telephone: (610) 647-1123 Fax: (610) 647-7705 Emergency Telephone: (800) 424-9300 Call Chemtrec
Prepared By	EHS&Q Coordinator Telephone: 1-319-235-1900

2. HAZARD(S) IDENTIFICATION

Classification: STOT RE1

Signal Word: **DANGER**

Hazard Statement: Causes damage to lungs through prolonged or repeated exposure via inhalation.

Precautionary statements: Do not breathe dust
In case of inadequate ventilation wear respiratory protection
Dispose of contents/containers in accordance with local regulation

Other: Long term exposure can cause lung injury (silicosis). IARC and NTP have determined that crystalline silica inhaled from occupational sources can cause cancer in humans. Risk of injury is dependent on the duration and level of exposure.

3. COMPOSITION/INFORMATION ON INGREDIENTS

SUBSTANCE	CAS Number	PERCENT	ACGIH TLV-TWA	OSHA PEL-TWA
Bentonite	1302-78-9	> 50%		Particulates Not Otherwise Regulated Respirable dust 5 mg/m^3 Total dust 15 mg/m^3
Coal Dust (Quartz) Respirable Fraction < 5% SiO_2	68409-95-0	< 40%	0.9 mg/m^3	2.4 mg/m^3
Cellulose Respirable Fraction Total Dust	9004-34-6	< 20%	10 mg/m^3	5 mg/m^3 15 mg/m^3
Gilsonite	12002-43-6	< 20%	5 mg/m^3	5 mg/m^3
Fire Clay Kaolinite (> 30%) Quartz (<30%)	1318-74-7 14808-60-7	< 12%		5 mg/m^3 $\frac{10\ mg/m^3}{\%SiO2 + 2}$
Graphite	7782-42-5	< 10%	2 mg/m^3	15 mppcf
Lignite Coal Dust (Quartz) Respirable Fraction < 5% SiO_2	68409-95-0	< 7%	0.9 mg/m^3	2.4 mg/m^3
Starch Respirable Fraction Total Dust	9005-28-8	< 2%	10 mg/m^3	5 mg/m^3 15 mg/m^3
Soda Ash	497-19-8	< 1%		
Crystalline Silica in the form of Quartz(Naturally occurring) Respirable dust Total dust	14808-60-7	1 – 5%	0.025 mg/m^3	$\frac{10\ mg/m^3}{\%SiO2 +2}$ $\frac{30\ mg/m^3}{\%SiO2 + 2}$
Crystalline silica, cristobalite (Naturally occurring) Respirable dust	14464-46-1	0 -1%	0.025 mg/m^3	1/2 x $\frac{10\ mg/m^3}{\%SiO2 + 2}$

More restrictive exposure limits may be enforced by some states, agencies, or other authorities.

4. FIRST AID MEASURES

Inhalation If inhaled, remove from area to fresh air. Get medical attention if respiratory irritation develops or if breathing becomes difficult.

Skin Wash with soap and water. Get medical attention if irritation persists.

Eyes In case of contact, immediately flush eyes with plenty of water for at least 15 minutes and get medical attention if irritation persists.

Ingestion	Do not induce vomiting. Slowly dilute with 1-2 glasses of water or milk and seek medical attention. Never give anything by mouth to an unconscious person.
Notes to Physician	Treat symptomatically.

5. FIRE FIGHTING MEASURES

Flash Point/Range (F):	Not Determined
Flash Point/Range (C):	Not Determined
Flash Point Method:	Not Determined
Autoignition Temperature (F):	Not Determined
Autoignition Temperature (C):	Not Determined
Flammability Limits in Air - Lower (%):	Not Determined
Flammability Limits in Air - Upper (%):	Not Determined

Fire Extinguishing Media	Water fog, carbon dioxide, foam, dry chemical.
Special Exposure Hazards	This product contains coal. At high temperatures toxic hydrocarbon gases may be formed; use with adequate ventilation to keep exposures below recommended limits. Good housekeeping practices are required to minimize this potential. Material becomes slippery when wet.
Special Protective Equipment for Fire-Fighters	Full protective clothing and approved self-contained breathing apparatus required for fire fighting personnel.
NFPA Ratings:	Health 1, Flammability 0, Reactivity 0
HMIS Ratings:	Health 1*, Flammability 0, Reactivity 0

6. ACCIDENTAL RELEASE MEASURES

Personal Precautionary Measures	Use appropriate protective equipment. Avoid creating and breathing dust.
Environmental Precautionary Measures	Prevent from entering sewers, waterways, or low areas.
Procedure for Cleaning / Absorption	Collect using dustless method and hold for appropriate disposal. Consider possible toxic or fire hazards associated with contaminating substances and use appropriate methods for collection, storage and disposal.

7. HANDLING AND STORAGE

Handling Precautions	This product contains quartz, which may become airborne without a visible cloud. Avoid breathing dust. Avoid creating dusty conditions. Use only with adequate ventilation to keep exposure below recommended exposure limits. Wear a NIOSH certified, European Standard En 149, or equivalent respirator when using this product. Wash hands after use. Material is slippery when wet.
Storage Information	Store away from oxidizers. Use good housekeeping in storage and work areas to prevent accumulation of dust. Close container when not in use. Do not reuse empty container.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Engineering Controls	Use approved industrial ventilation and local exhaust as required to maintain exposures below applicable exposure limits listed in Section 3.
Respiratory Protection	Wear a NIOSH certified, European Standard EN 149, or equivalent respirator when using this product.
Hand Protection	Impervious rubber gloves.
Skin Protection	Wear clothing appropriate for the work environment. Dusty clothing should be laundered before reuse. Use precautionary measures to avoid creating dust when removing or laundering clothing.
Eye Protection	Wear safety glasses or goggles to protect against exposure.
Other Precautions	None known.

9. PHYSICAL AND CHEMICAL PROPERTIES

Physical State:	Solid
Color:	Light Gray to Dark Gray
Odor:	Odorless
pH:	7 - 10
Specific Gravity @ 20 C (Water=1):	1.7 – 2.65
Density @ 20 C (lbs./gallon):	Not Determined
Bulk Density @ 20 C (lbs/ft3):	44 - 58
Boiling Point/Range (F):	Not Determined
Boiling Point/Range (C):	Not Determined
Freezing Point/Range (F):	Not Determined
Freezing Point/Range (C):	Not Determined
Vapor Pressure @ 20 C (mmHg):	Not Determined
Vapor Density (Air=1):	Not Determined
Percent Volatiles:	Not Determined
Evaporation Rate (Butyl Acetate=1):	Not Determined
Solubility in Water (g/100ml):	Insoluble
Solubility in Solvents (g/100ml):	Not Determined
VOCs (lbs./gallon):	Not Determined
Viscosity, Dynamic @ 20 C (centipoise):	Not Determined
Viscosity, Kinematic @ 20 C (centistrokes):	Not Determined
Partition Coefficient/n-Octanol/Water:	Not Determined
Molecular Weight (g/mole):	Not Determined

10. STABILITY AND REACTIVITY

Stability Data:	Stable
Hazardous Polymerization:	Will Not Occur
Conditions to Avoid	None anticipated
Incompatibility (Materials to Avoid)	Strong oxidizers.
Hazardous Decomposition Products	Various oxides of Carbon, and possibly sulfur dioxide. At elevated temperatures toxic hydrocarbons may be formed.

Additional Guidelines Not Applicable

11. TOXICOLOGICAL INFORMATION

Principle Rout of Exposure Eye or skin contact, inhalation

Inhalation Inhaled crystalline silica in the form of quartz or cristobalite from occupational sources is carcinogenic to humans (IARC, Group1). There is sufficient evidence in experimental animals for the carcinogenicity of tridymite (IARC Group 2A).

Skin Contact May cause mechanical skin irritation.

Eye Contact May cause eye irritation.

Ingestion None Known

Aggravated Medical Conditions Individuals with respiratory disease, including but not limited to asthma and bronchitis, or subject to eye irritation, should not be exposed to quartz dust.

Chronic Effects/Carcinogenicity Cancer Status In 2011, The International Agency for Research on Cancer (IARC) concluded that crystalline silica in the form of quartz or cristobalite dust is carcinogenic to humans (IARC Group 1). Prolonged inhalation of silica dust may result in silicosis, a progressive, disabling, and sometimes fatal, lung disease. Avoid creating dusty conditions and use a NIOSH/MSHA approved dust respirator when Threshold limit Values (TLVs) may be exceeded.

Other Information

Toxicity Tests

Oral Toxicity:	Not determined
Dermal Toxicity:	Not determined
Inhalation Toxicity:	Not determined
Primary Irritation Effect:	Not determined
Carcinogenicity	Refer to IARC Monograph 100 volume C (2011).
Genotoxicity:	Not determined
Reproductive / Developmental Toxicity:	Not determined

12. ECOLOGICAL INFORMATION

Mobility (Water/Soil/Air) Not determined

Persistence/Degradability Not determined

Bio-accumulation Not determined

Ecotoxicological Information

Acute Fish Toxicity:	Not determined
Acute Crustaceans Toxicity:	Not determined
Acute Algae Toxicity:	Not determined

Chemical Fate Information	Not determined
Other Information	Not applicable

13. DISPOSAL CONSIDERATIONS

Disposal Method	Disposal should be made in accordance with federal, state, and local regulations.
Contaminated Packaging	Follow all applicable national or local regulations.

14. TRANSPORT INFORMATION

Land Transportation

DOT
Not restricted

Canadian TDG
Not restricted

ADR Not restricted

Air Transportation
ICAO/IATA Not restricted

Sea Transportation
IMDG Not restricted

Other Shipping Information

Labels: None

15. REGULATORY INFORMATION

US Regulations

US TSCA Inventory	All components listed on inventory
EPA SARA Title III Extremely Hazardous Substances	Not applicable
EPA SARA (311,312) Hazard Class	Acute Health Hazard Chronic Health Hazard
EPA SARA (313) Chemicals	This product does not contain a toxic chemical for routine annual "Toxic Chemical Release Reporting" under Section 313 (40 CFR 372).

EPA CERCLA/Superfund Reportable Spill Quantity For This Product	Not Applicable.
EPA RCRA Hazardous Waste Classification	If product becomes a waste, it does NOT meet the criteria of a hazardous waste as defined by the US EPA.
California Proposition 65	The California Proposition 65 regulations apply to this product
MA Right-to-Know Law	One or more components listed.
NJ Right-to-Know Law	One or more components listed.
PA Right-to-Know Law	One or more components listed.
Canadian Regulations	
Canadian DSL Inventory	Product contains one or more components not listed on inventory.
WHMIS Hazard Class	D2A Very Toxic Materials Crystalline silica

16. OTHER INFORMATION

The following sections have been revised since the last issue of this SDS
Sections 1,2,11,16.

Additional Information

For additional information on the use of this product, contact your local Metalcasting Performance Additives representative.

For questions about the Safety Data Sheet for this or other Metalcasting Performance Additives products, contact EHS&Q Coordinator at 1-319-235-1900.

Disclaimer Statement

As of the date of preparation of this document, the forgoing information is believed to be accurate and is provided in good faith to comply with applicable federal and state law(s). This information is furnished without warranty, expressed or implied, as to accuracy or completeness. The information is obtained from various sources including the manufacturer and other third party sources. The information may not be valid under all conditions nor if this material is used in combination with other materials or in any process. Final determination of suitability of any material is the sole responsibility of the user.

*****END OF SDS*****